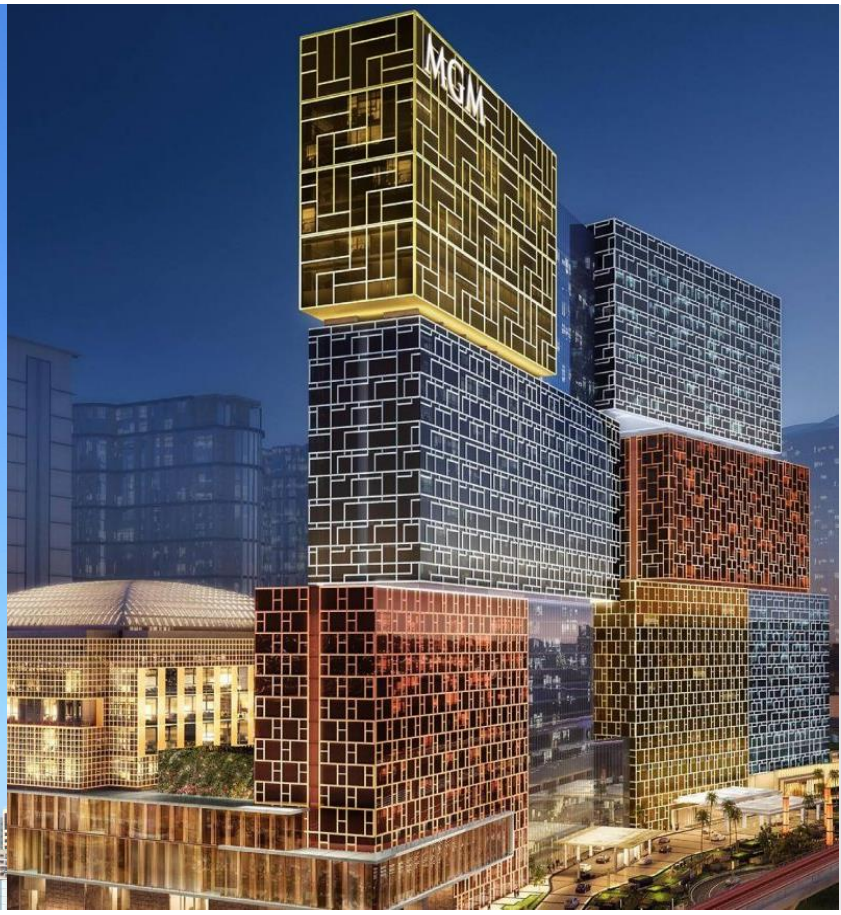
MGM MACAU & COTAI
RESOURCE EFFICIENCY MANAGEMENT REPORT
MAY 2023

Prepared For :

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EXECUTIVE SUMMARY

NV5 has been working closely with MGM to implement the EMOs as well as identify additional opportunities to achieve ISO certification energy targets. NV5 appreciates the great support and trust from MGM team over the years, in particular the Facility Management and the Sustainability Teams.

Key Performance Indicators

NV5 reports both normalized and actual savings on a monthly basis for both properties. The normalized savings are determined based on a 2019 baseline, with aligned annual savings targets to meet both ISO50001 as well as NV5's retainer contract requirements. Please note that the reported normalized savings achieved are to-date and cumulative for each month of the respective year.

Table 1. ISO and Retainer Contract Energy Performance Targets for 2025

MGM Properties	Baseline	Energy Savings	2021	2022	Normalized Savings Achieved Up to May 2023	2024	2025
MACAU	2019	Target	-4%	-8%	-10%	N/A	**-13%**
		Actual	**-5.9%**	**-10.8%**	**-6.2%**	-	**-**
COTAI	2019	Target	-4%	-8%	-10%	N/A	**-13%**
		Actual	**-6.4%**	**-16.9%**	**-7.8%**	-	**-**

The actual monthly savings are determined against the 2019 baseline consumption and cumulative for each month of 2023.

Table 2. Internal Actual Energy Performance Targets for 2025

MGM Properties	Baseline	Actual Saving Target[1]	Savings Achieved Up to May 2023
MACAU	2019	13%	**-4.5%**
COTAI	2019	13%	**-7.6%**

[1] Actual Energy Savings Target for 2025 has been established based on future CAPEX projects, following review with MGM teams.

Highlighted Changes in Operation

MACAU

1. New water-to-water HX was installed for podium MVAC space heating, testing works were conducted since Feb 2023. Continuous testing was being performed.
2. Grande Praça 7 Units AHU were installed with PICCV.

COTAI

1. Grill 58, MDM, Ballroom, Hao Guo kitchen exhaust fan steps speed control program and graphics were updated in BMS.
2. MGM COTAI hotel towers façade lighting control has transferred to IT team, it will be dimmed down from 02:00 – 17:30 instead of existing switch OFF. This is to reduce the LED failure rate and extend the LED life time.
3. MGM COTAI is planning to install additional EV chargers in B2/B3 car park.

Consumption Intensity

The 2023 YTD electricity and water use intensities are tracked on a monthly basis for both properties:

Table 3. Year to Date Actual Electricity Use Intensity (EUI)

MGM Properties	YTD EUI (kWh/m²)	% Change Compared to 2022
MACAU	**24.3**	**+3.7%**
COTAI	**24.6**	**+6.5%**

Table 4. Year to Date Actual Water Use Intensity (WUI)

MGM Properties	YTD WUI (m³/ m²)	% Change Compared to 2022
MACAU	**0.22**	**+11.8%**
COTAI	**0.18**	**+35.5%**

ISO 50001 Certification Audit

MACAU

The HKQAA 2023 surveillance audit was completed on 28 April 2023. There is no nonconformity (NC) and areas for improvement (AFI) from auditors.

COTAI

The HKQAA 2022 recertification audit was completed on 30 September 2022. There is no nonconformity (NC) and areas for improvement (AFI) from auditors.

Target Milestones in Year 2023

The following activities are to be completed in year 2023:

Table 5. MGM MACAU Year 2023 Projects

Item	Initiatives	Status
Electricity		
1	Secondary Chilled Water Pump Retrofit (Phase 1)	On-going
2	Cooling Tower Fan Retrofit (Phase 1)	On-going
3	PICCV Implementation 2023 (Grande Praça 7 Units)	Completed
4	Neon Light LED Replacement (Phase 3)	On-going
LPG		
5	Natural Gas Changeover Project	On-going

Table 6. MGM COTAI Year 2023 Projects

Item	Initiatives	Status
Electricity		
1	Chiller Plant Controls Optimization (CHWP DP Setpoint Reset, CHWST Setpoint Reset)	On-going
2	PICCV Implementation 2023	On-going
3	KEF Steps Speed Control Optimization (Hao Guo, Five Foot Road, Chún, Coast)	On-going
4	AHU Static Pressure & Supply Air Temperature Set-Point Reset (Phase 2)	On-going
5	Centralized RO Pump Set Replacement (10 units)	Completed
6	Booster Pump Set Replacement (2 units)	On-going
Water		
7	Extension of B2 Carpark CDP Water Recycling (Phase 5 to 7)	On-going
8	Spectacle Façade Rainwater Harvesting for CT Make Up	On-going

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ACKNOWLEDGMENT

NV5 has been working with MGM Team as the energy and sustainability manager to achieve energy savings in MGM properties and feed operational intelligence into the design.

The major goal of this project is not only to realize operation energy savings but also to extend the life of equipment, hence reducing CAPEX. MGM has demonstrated continuous commitment to be an environmentally and socially responsible business in the past years. In addition to that NV5 has assisted MGM MACAU and COTAI to achieve several other energy management and sustainability awards and certifications, such as ISO 50001, CEM Energy Savings Award, Macao Green Hotel Award and China 3 Star Green Building Operation Label.

NV5 appreciates the great support and trust from MGM Team over the years, especially the Facilities Management and the Sustainability Teams.

1.0 PORTFOLIO PERFORMANCE SUMMARY

1.1 ELECTRICITY USE INTENSITY (EUI)

MACAU's electricity use intensity (EUI) in May 2023 had increased by 9.9% compared to the same month in 2022 and increased by 3.7% in 2023 compared to 2022. Meanwhile, COTAI's EUI in May 2023 had increased by 14.7% compared to the same month in 2022 and increased by 6.5% in 2023 compared to 2022.

Table 7. Actual Electricity Consumption Intensity (EUI) Comparison

Monthly Period	MACAU			COTAI		
	Year 2022 Actual EUI (kWh/m²)	Year 2023 Actual EUI (kWh/m²)	% Change	Year 2022 Actual EUI (kWh/m²)	Year 2023 Actual EUI (kWh/m²)	% Change
January	22.6	22.4	-0.8%	22.3	22.6	+1.6%
February	20.2	21.2	+5.2%	20.1	21.0	+4.5%
March	24.4	24.6	+1.0%	23.6	24.4	+3.3%
April	24.3	25.0	+3.0%	24.0	25.8	+7.3%
May	25.9	28.5	+9.9%	25.6	29.4	+14.7%
June						
July						
August						
September						
October						
November						
December						
Average	**23.5**	**24.3**	**+3.7%**	**23.1**	**24.6**	**+6.5%**

1.2 WATER USE INTENSITY (WUI)

MACAU's water use intensity (WUI) in May 2023 increased by 20.1% compared to the same month in 2022 and increased by 11.8% in 2023 compared to 2022. Meanwhile, COTAI's WUI in May 2023 increased by 44.2% compared to the same month in 2022 and increased by 35.5% for 2023 compared to 2022.

Table 8. Actual Water Consumption Intensity (WUI) Comparison

Monthly Period	MACAU			COTAI		
	Year 2022 Actual WUI (m^3/m^2)	Year 2023 Actual WUI (m^3/m^2)	% Change	Year 2022 Actual WUI (m^3/m^2)	Year 2023 Actual WUI (m^3/m^2)	% Change
January	0.18	0.19	+6.0%	0.12	0.14	+21.0%
February	0.20	0.21	+8.6%	0.12	0.16	+35.4%
March	0.19	0.21	+11.9%	0.12	0.17	+37.9%
April	0.21	0.24	+10.3%	0.14	0.19	+36.7%
May	0.23	0.28	+20.1%	0.15	0.21	+44.2%
June						
July						
August						
September						
October						
November						
December						
Average	**0.20**	**0.22**	**+11.8%**	**0.13**	**0.18**	**+35.5%**

2.0 MGM MACAU PERFORMANCE

2.1 ELECTRICITY CONSUMPTION

The actual electricity consumption and cost in May 2023 increased by **9.9%** and increased by **7.8%** respectively compared to the same month in 2022. To date, MGM MACAU has increased the actual electricity consumption by **3.7%** and associated cost increased by **4.8%** respectively.

Table 9. MACAU Summary of Actual Monthly Electricity Consumption and Cost

Monthly Period	Actual Electricity Consumption (kWh)			Actual Electricity Cost (MOP)			% Change – CDD
	2022	2023	% Change Actual	2022	2023	% Change Actual	
January	4,784,280	4,745,400	-0.8%	5,519,350	5,668,020	+2.7%	-37.7%
February	4,266,240	4,488,360	+5.2%	4,942,590	5,365,140	+8.5%	+658.0%
March	5,152,320	5,205,240	+1.0%	5,938,160	6,207,870	+4.5%	-9.0%
April	5,131,680	5,284,800	+3.0%	6,183,420	6,236,780	+0.9%	-3.2%
May	5,478,840	6,019,800	+9.9%	6,595,030	7,106,470	+7.8%	+17.6%
June							
July							
August							
September							
October							
November							
December							
Total/Average	**24,813,360**	**25,743,600**	**+3.7%**	**29,178,550**	**30,584,280**	**+4.8%**	**+9.6%**

***Important note**: Cooling Degree Days (CDDs) is the main variable taken into consideration on electricity consumption normalization.*

NV5 summarized the key items which could result in electricity consumption increase below:

- Chiller plant electricity consumption increased by 10.0% in May 2023 compared with the same month in 2022 because the CDD has increased by 17.6%
- Heat pump plant electricity consumption has increased by 51.4% in May 2023 compared with the same month in 2022 because heat pumps were utilized for the dehumidification instead of LPG boilers.
- Hotel tower consumption has increased by 23.8% in May 2023 compared with the same month in 2022 with full operation occupancy.

Table 10 presents the comparison of electricity rates in the years 2022 and 2023. The electricity rate in May 2023 decreased by 1.9% compared to the same month in 2022.

Table 10. MACAU - Summary of Actual Monthly Electricity Rate

Monthly Period	Year 2022 – Electricity Rate (MOP/kWh)	Year 2023 – Electricity Rate (MOP/kWh)	% Change – Electricity Rate
January	1.15	1.19	+3.5%
February	1.16	1.20	+3.2%
March	1.15	1.19	+3.5%
April	1.20	1.18	-2.1%
May	1.20	1.18	-1.9%
June			
July			
August			
September			
October			
November			
December			
Total/Average	**1.17**	**1.19**	**+1.2%**

Table 11 summarizes the monthly normalized electricity consumption. The actual electricity consumption in May 2023 decreased by 2.4% compared to 2019 normalized consumption in the same month.

Table 11. MACAU - Summary of Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2023 Actual Electricity Consumption (kWh)	% Change – Normalized
January	4,715,548	4,745,400	+0.6%
February	4,893,297	4,488,360	-8.3%
March	5,275,299	5,205,240	-1.3%
April	5,597,956	5,284,800	-5.6%
May	6,166,350	6,019,800	-2.4%
June			
July			
August			
September			
October			
November			
December			
Total/Average	**26,648,451**	**25,743,600**	**-3.4%**

2.2 LPG CONSUMPTION

LPG consumption can be broken down into three (3) categories: boiler, hotel upper zone DHW, and kitchens/restaurants. Since the boiler and restaurants LPG utilization cover over 95% of the overall site LPG consumption, NV5 focuses the analysis on those two areas.

According to Table 12, the actual LPG consumption in May 2023 decreased by **2.8%** and cost decreased **15.4%** respectively compared to the same month in 2022. To date, MGM MACAU has decreased the actual LPG consumption by **17.2%** and associated costs decreased by **23.9%.**

Table 12. MACAU – Summary of Actual Monthly LPG Consumption and Cost

Monthly Period	Actual LPG Consumption (kg)			Actual LPG Cost (MOP)			% Change – Min OAT
	2022	2023	% Change Actual	2022	2023	% Change Actual	
January	121,221	123,108	+1.6%	1,327,490	1,084,579	-18.3%	-13.7%
February	146,623	78,483	-46.5%	1,450,103	768,347	-47.0%	+105.6%
March	91,040	77,597	-14.8%	900,382	759,672	-15.6%	+5.7%
April	85,743	77,139	-10.0%	858,756	755,191	-12.1%	+42.3%
May	82,704	80,383	-2.8%	992,447	839,195	-15.4%	+15.4%
June							
July							
August							
September							
October							
November							
December							
Total/Average	**527,330**	**436,709**	**-17.2%**	**5,529,178**	**4,206,984**	**-23.9%**	**+23.4%**

Important note: *Minimum Outdoor Air Temperature (OAT) is the main variable taken into consideration on LPG consumption normalization.*

NV5 summarized the key items which could result in LPG consumption decrease below:

- LPG Boiler consumption decreased by 25.1% in May 2023 compared with the same month in 2022 due to the utilization of the new water-to-water HX for space heating instead of utilizing the LPG boiler.

Table 13 presents the comparison of LPG rates in the years 2022 and 2023. The LPG rate in May 2023 decreased by 13.0% compared with the same month in 2022.

Table 13. MACAU - Summary of Actual Monthly LPG Rate

Monthly Period	Year 2022 – LPG Rate (MOP/kg)	Year 2023 – LPG Rate (MOP/kg)	% Change – LPG Rate
January	10.95	8.81	-19.6%
February	9.89	9.79	-1.0%
March	9.89	9.79	-1.0%
April	10.02	9.79	-2.3%
May	12.00	10.44	-13.0%
June			
July			
August			
September			
October			
November			
December			
Total/Average	**10.55**	**9.72**	**-7.8%**

Table 14 summarizes the monthly normalized LPG consumption. The actual LPG consumption in May 2023 decreased by 0.6% compared to 2019 normalized consumption in the same month.

Table 14. MACAU - Summary of Normalized Monthly LPG Consumption

Monthly Period	2019 Normalized LPG Consumption (kg)	2023 Actual LPG Consumption (kg)	% Change – Normalized
January	103,641	123,108	+18.8%
February	84,783	78,483	-7.4%
March	77,414	77,597	+0.2%
April	76,016	77,139	+1.5%
May	80,862	80,383	-0.6%
June			
July			
August			
September			
October			
November			
December			
Total/Average	**422,716**	**436,709**	**+3.3%**

2.3 ISO50001 ENERGY CONSUMPTION RECORD

The MACAU ISO50001 energy performance target is to achieve 13% normalized energy savings by 2025 compared to the 2019 baseline. To date, normalized energy saving achieved up to May 2023 is **6.2%** as shown in Table 15. Table 16 below summarizes the site monthly energy consumption (Electricity and LPG combined) compared to the 2019 baseline.

Table 15. MACAU ISO 50001 Normalized Energy Performance Targets for 2025

MGM Property	Baseline	Energy Savings	2021	2022	Normalized Savings Achieved Up to May 2023	2024	2025
MACAU	2019	Target	-4%	-8%	-10%	N/A	**-13%**
		Actual	**-5.9%**	**-10.8%**	**-6.2%**	-	**-**

Table 16. MACAU ISO 50001 Normalized Monthly Energy Consumption

Monthly Period	2019 Normalized Energy Consumption (GJ)	2023 Actual Energy Consumption (GJ)	% Change – Normalized
January	23,023	22,660	-1.6%
February	23,041	19,713	-14.4%
March	22,946	22,254	-3.0%
April	25,252	22,520	-10.8%
May	25,652	25,313	-1.3%
June			
July			
August			
September			
October			
November			
December			
Total/Average	**119,914**	**112,460**	**-6.2%**

Important note: *Factors used for unit conversion: Electricity (0.0036) and LPG (0.0453)*

2.4 WATER CONSUMPTION

Based on the existing water metering system, the overall site water consumption can be broken down into five (5) categories: Kitchen and Restaurants; Hotel Tower; Fire Services; Chiller Plant & miscellaneous.

According to Table 17, the actual water consumption and cost in May 2023 has increased by 20.1% and 20.0% compared to the same month in 2022 respectively. To date, MGM MACAU has increased both the actual water consumption and associated cost by 11.8%.

Table 17. MACAU - Summary of Actual Monthly Water Consumption and Cost

Monthly Period	Actual Water Consumption (m^3)			Actual Water Cost (MOP)			Normalization Factors	
	2022	2023	% Change Actual	2022	2023	% Change Actual Cost	% Change – CDD	% Change – Food Covers
January	37,078	39,305	+6.0%	287,500	305,020	+6.1%	-37.7%	+52.7%
February	41,524	45,112	+8.6%	322,270	350,080	+8.6%	+658.0%	+16.6%
March	39,962	44,725	+11.9%	310,160	347,050	+11.9%	-9.0%	+71.8%
April	45,295	49,960	+10.3%	351,540	387,650	+10.3%	-3.2%	+87.5%
May	49,042	58,878	+20.1%	380,590	456,800	+20.0%	+17.6%	+77.2%
June								
July								
August								
September								
October								
November								
December								
Total/Average	**212,901**	**237,980**	**+11.8%**	**1,652,060**	**1,846,600**	**+11.8%**	**+9.6%**	**+58.0%**

Important note: *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on water consumption normalization.*

NV5 summarized the key items which could result in water consumption increase below:

- Hot water consumption for the hotel and for the kitchen for May 2023 has increased by 15.0% and 34.4% respectively compared to the same month in 2022 due to the 85.4% and 77.2% increase of occupants and food cover.

Table 18 presents the comparison of water rates in the years 2022 and 2023. The water rate in May 2023 has decreased by 0.03% compared to the same month in 2022. The average utility rate of water remains unchanged compared to 2022.

Table 18. MACAU - Summary of Actual Monthly Water Rates

Monthly Period	Year 2022 – Water Rate (MOP/m³)	Year 2023 – Water Rate (MOP/m³)	% Change – Water Rate
January	7.754	7.760	+0.1%
February	7.761	7.760	-0.01%
March	7.761	7.760	-0.02%
April	7.761	7.759	-0.02%
May	7.760	7.758	-0.03%
June			
July			
August			
September			
October			
November			
December			
Total/Average	**7.760**	**7.760**	**-**

Table 19 summarized the monthly normalized water consumption. The actual water consumption in May 2023 increased by 6.2% compared to 2019 normalized consumption in the same month.

Table 19. MACAU - Summary of Normalized Monthly Water Consumption

Monthly Period	2019 Normalized Water Consumption (m³)	2023 Actual Water Consumption (m³)	% Change – Normalized
January	43,437	39,305	-9.5%
February	42,590	45,112	+5.9%
March	47,450	44,725	-5.7%
April	50,917	49,960	-1.9%
May	55,459	58,878	+6.2%
June			
July			
August			
September			
October			
November			
December			
Total/Average	**239,853**	**237,980**	**-0.8%**

2.5 MACAU YEAR 2023 PROJECTS

NV5 will continue to perform analysis and coordinate with MGM to implement the following projects in 2023 and reduce the site energy and water consumption which supports NV5 annual normalized energy savings target.

Table 20. MGM MACAU Year 2023 Projects

Item	Initiatives	Status
Electricity		
1	Secondary Chilled Water Pump Retrofit (Phase 1)	On-going
2	Cooling Tower Fan Retrofit (Phase 1)	On-going
3	PICCV Implementation 2023 (Grande Praça 7 Units)	Completed
4	Neon Light LED Replacement (Phase 3)	On-going
LPG		
5	Natural Gas Changeover Project	On-going

3.0 MGM COTAI PERFORMANCE

3.1 ELECTRICITY CONSUMPTION

The actual electricity consumption and cost in May 2023 increased by 14.7% and increased by 12.9% respectively compared to the same month in 2022. To date, MGM COTAI has increased the actual electricity consumption by 6.5% and associated cost increased by 7.7%.

Table 21. COTAI Summary of Actual Monthly Electricity Consumption and Cost

Monthly Period	Actual Electricity Consumption (kWh)			Actual Electricity Cost (MOP)			% Change – CDD
	2022	2023	% Change Actual	2022	2023	% Change Actual	
January	8,336,400	8,471,280	+1.6%	10,013,460	10,516,100	+5.0%	-37.7%
February	7,528,560	7,869,600	+4.5%	9,066,060	9,794,150	+8.0%	+658.0%
March	8,833,200	9,128,520	+3.3%	10,632,040	11,334,370	+6.6%	-9.0%
April	8,985,360	9,638,280	+7.3%	11,262,370	11,855,350	+5.3%	-3.2%
May	9,585,000	10,993,680	+14.7%	11,983,070	13,528,490	+12.9%	+17.6%
June							
July							
August							
September							
October							
November							
December							
Total/Average	**43,268,520**	**46,101,360**	**+6.5%**	**52,957,000**	**57,028,460**	**+7.7%**	**+9.6%**

***Important note**: Cooling Degree Days (CDDs) is the main variable taken into consideration on electricity consumption normalization*

NV5 summarized the key items which could result in electricity consumption increase below:

- Hotel Towers electricity consumption for May 2023 has increased by 27.6% compared to the same month in 2022 due to the significant increase of occupancy.
- Chiller plant electricity consumption for May 2023 has increased by 34.1% compared to the same month in 2022 due to the significant increase of CDD.

Table 22 presents the monthly e-bus electricity consumption in MGM COTAI. The e-bus electricity consumption in May 2023 has increased by 158.6% compared to the same month in 2022.

Table 22. COTAI - Summary of Monthly e-bus Electricity Consumption

Monthly Period	E-bus Electricity Consumption (kWh)		
	2022	2023	% Change – Actual
January	42,287	51,261	+21.2%
February	47,905	53,640	+12.0%
March	38,793	66,865	+72.4%
April	38,537	72,669	+88.6%
May	51,328	132,741	+158.6%
June			
July			
August			
September			
October			
November			
December			
Total/Average	**218,848**	**377,176**	**+72.3%**

Table 23 presents the comparison of electricity rates in the years 2022 and 2023. The electricity rate in May 2023 decreased by 1.6% compared to the same month in 2022.

Table 23. COTAI - Summary of Actual Monthly Electricity Rate

Monthly Period	Year 2022 – Electricity Rate (MOP/kWh)	Year 2023 – Electricity Rate (MOP/kWh)	% Change – Electricity Rate
January	1.20	1.24	+3.3%
February	1.20	1.24	+3.3%
March	1.20	1.24	+3.2%
April	1.25	1.23	-1.9%
May	1.25	1.23	-1.6%
June			
July			
August			
September			
October			
November			
December			
Total/Average	**1.22**	**1.24**	**+1.2%**

Table 24 summarizes the monthly normalized electricity consumption. The actual electricity consumption in May 2023 increased by 2.5% compared to 2019 normalized consumption in the same month.

Table 24. COTAI - Summary of Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2023 Actual Electricity Consumption (kWh)	% Change – Normalized
January	8,521,687	8,471,280	-0.6%
February	8,791,529	7,869,600	-10.5%
March	9,371,451	9,128,520	-2.6%
April	9,861,279	9,638,280	-2.3%
May	10,724,164	10,993,680	+2.5%
June			
July			
August			
September			
October			
November			
December			
Total/Average	**47,270,110**	**46,101,360**	**-2.5%**

3.2 NATURAL GAS CONSUMPTION

The actual natural gas consumption and cost in May 2023 increased by 45.0% and 58.2% respectively compared to the same month in 2022, respectively. To date, MGM COTAI has increased actual natural gas consumption by 13.4% and associated costs by 20.4%.

Table 25. COTAI - Summary of Actual Monthly Natural Gas Consumption and Cost

Monthly Period	Actual NG Consumption (m^3)			Actual NG Cost (MOP)			% Change – Min OAT	% Change – Food Covers
	2022	2023	% Change Actual	2022	2023	% Change Actual		
January	59,378	64,043	+7.9%	376,240	405,800	+7.9%	-13.7%	+67.3%
February	76,089	58,053	-23.7%	482,130	378,840	-21.4%	+105.6%	+29.8%
March	52,146	63,461	+21.7%	330,410	438,660	+32.8%	+5.7%	+135.0%
April	43,900	62,658	+42.7%	278,170	433,120	+55.7%	+42.3%	+184.5%
May	45,632	66,163	+45.0%	289,140	457,350	+58.2%	+15.4%	+197.2%
June								
July								
August								
September								
October								
November								
December								
Total/Average	**277,145**	**314,378**	**+13.4%**	**1,756,090**	**2,113,770**	**+20.4%**	**+23.4%**	**+106.6%**

Important note: *Minimum Outdoor Air Temperature (OAT is the main variable taken into consideration on natural gas consumption normalization*

NV5 summarized the key items which could result in natural gas consumption increase below:

- Kitchen NG consumption for May 2023 has increased by 44.3% compared to the same month in 2022.

Table 26 presents the comparison of natural gas rates in the years 2022 and 2023. The natural gas rate in May 2023 increased by 9.1% compared to the same month in 2022.

Table 26. COTAI - Summary of Actual Monthly Natural Gas Rate

Monthly Period	Year 2022 – Natural Gas Rate (MOP/m³)	Year 2023 – Natural Gas Rate (MOP/m³)	% Change – NG Rate
January	6.34	6.34	-
February	6.34	6.53	+3.0%
March	6.34	6.91	+9.1%
April	6.34	6.91	+9.1%
May	6.34	6.91	+9.1%
June			
July			
August			
September			
October			
November			
December			
Total/Average	**6.34**	**6.72**	**+6.1%**

Table 27 summarizes the monthly normalized natural gas consumption. The actual natural gas consumption in May 2023 decreased by 41.1% compared to 2019 normalized consumption in the same month.

Table 27. COTAI - Summary of Normalized Monthly Natural Gas Consumption

Monthly Period	2019 Normalized Natural Gas Consumption (m³)	2023 Actual Natural Gas Consumption (m³)	% Change – Normalized
January	128,174	64,043	-50.0%
February	120,801	58,053	-51.9%
March	117,977	63,461	-46.2%
April	113,428	62,658	-44.8%
May	112,330	66,163	-41.1%
June			
July			
August			
September			
October			
November			
December			
Total/Average	**592,710**	**314,378**	**-47.0%**

3.3 ISO50001 ENERGY CONSUMPTION RECORD

COTAI ISO50001 energy target is to achieve 13% normalized energy consumption reduction by 2025 compared to the 2019 baseline. To date, normalized energy saving achieved up to May 2023 is **7.8%** as shown in Table 28 and Table 29 below summarizes the site monthly energy consumption (Electricity and Natural Gas combined) compared to the 2019 baseline.

Table 28. COTAI ISO 50001 Normalized Energy Performance Targets for 2025

MGM Property	Baseline	Energy Savings	2021	2022	Normalized Savings Achieved Up to May 2023	2024	2025
COTAI	2019	Target	-4%	-8%	-10%	N/A	**-13%**
		Actual	**-6.4%**	**-16.9%**	**-7.8%**	-	**-**

Table 29. COTAI ISO 50001 Normalized Monthly Energy Consumption

Monthly Period	2019 Normalized Energy Consumption (GJ)	2023 Actual Energy Consumption (GJ)	% Change – Normalized
January	35,567	32,719	-8.0%
February	36,708	30,345	-17.3%
March	37,322	35,065	-6.0%
April	40,715	36,872	-9.4%
May	41,466	41,873	+1.0%
June			
July			
August			
September			
October			
November			
December			
Total/Average	**191,778**	**176,874**	**-7.8%**

***Important note**: Factors used for unit conversion: Electricity (0.0036) and Natural Gas (0.0347)*

3.4 WATER CONSUMPTION

According to Table 30, the actual water consumption and associated costs in May 2023 increased by **44.2%** and **44.0%**, respectively compared to the same month in 2022. To date, MGM COTAI has increased actual water consumption by **35.5%**, and associated costs by **35.4%**.

Table 30. COTAI - Summary of Actual Monthly Water Consumption and Cost

Monthly Period	Actual Water Consumption (m^3)			Actual Water Cost (MOP)			Normalization Factors	
	2022	2023	% Change Actual	2022	2023	% Change Actual Cost	% Change – CDD	% Change – Food Covers
January	44,658	54,018	+21.0%	347,240	419,790	+20.9%	-37.7%	+67.3%
February	45,298	61,317	+35.4%	352,330	476,420	+35.2%	+658.0%	+29.8%
March	44,914	61,936	+37.9%	349,260	481,140	+37.8%	-9.0%	+135.0%
April	51,640	70,617	+36.7%	401,500	548,440	+36.6%	-3.2%	+184.5%
May	55,628	80,211	+44.2%	432,480	622,920	+44.0%	+17.6%	+197.2%
June								
July								
August								
September								
October								
November								
December								
Total/Average	**242,138**	**328,099**	**+35.5%**	**1,882,810**	**2,548,710**	**+35.4%**	**+9.6%**	**+106.6%**

Important note: *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization*

NV5 summarized the key items which could result in water consumption increased below:

- Kitchen water consumption for May 2023 has increased by 31.1% compared with the same month last year.
- Hotel Towers water consumption for May 2023 has increased by 275.6% compared with the same month last year.
- Cooling Tower make up water consumption for May 2023 has increased by 41.4% compared with the same month last year

Table 31 presents the comparison of water rates in the years 2022 and 2023. The water rate in May 2023 decreased by 0.1% compared to the same month in 2022. To date, the average utility rate of water has decreased by 0.1% compared to 2022.

Table 31. COTAI - Summary of Actual Monthly Water Rates

Monthly Period	Year 2022 – Water Rate (MOP/m³)	Year 2023 – Water Rate (MOP/m³)	% Change – Water Rate
January	7.776	7.771	-0.05%
February	7.778	7.770	-0.1%
March	7.776	7.768	-0.1%
April	7.775	7.766	-0.1%
May	7.775	7.766	-0.1%
June			
July			
August			
September			
October			
November			
December			
Total/Average	**7.776**	**7.768**	**-0.1%**

Table 32 summarized the monthly normalized water consumption. The actual water consumption in May 2023 increased by 5.6% compared to 2019 normalized consumption in the same month.

Table 32. COTAI - Summary of Normalized Monthly Water Consumption

Monthly Period	2019 Normalized Water Consumption (m³)	2023 Actual Water Consumption (m³)	% Change – Normalized
January	57,914	54,018	-6.7%
February	56,894	61,317	+7.8%
March	62,913	61,936	-1.6%
April	69,294	70,617	+1.9%
May	75,963	80,211	+5.6%
June			
July			
August			
September			
October			
November			
December			
Total/Average	**322,977**	**328,099**	**+1.6%**

3.5 COTAI YEAR 2023 PROJECTS

NV5 will continue to perform analysis and coordinate with MGM to implement the following projects in 2023 and reduce the site energy and water consumption which supports NV5 annual normalized energy savings target.

Table 33. MGM COTAI Year 2023 Projects

Item	Initiatives	Status
Electricity		
1	Chiller Plant Controls Optimization (CHWP DP Setpoint Reset, CHWST Setpoint Reset)	On-going
2	PICCV Implementation 2023	On-going
3	KEF Steps Speed Control Optimization (Hao Guo, Five Foot Road, Chún, Coast)	On-going
4	AHU Static Pressure & Supply Air Temperature Set-Point Reset (Phase 2)	On-going
5	Centralized RO Pump Set Replacement (10 units)	Completed
6	Booster Pump Set Replacement (2 units)	On-going
Water		
7	Extension of B2 Carpark CDP Water Recycling (Phase 5 to 7)	On-going
8	Spectacle Façade Rainwater Harvesting for CT Make Up	On-going

APPENDIX A

Regression Analysis

The 2019 normalized consumption modelling is remained used for tracking the agreed normalized energy savings target for MGM MACAU and MGM COTAI. The following tables show a summary of the selected independent variables for each utility which are used for the normalized consumption modelling in 2023.

MGM MACAU

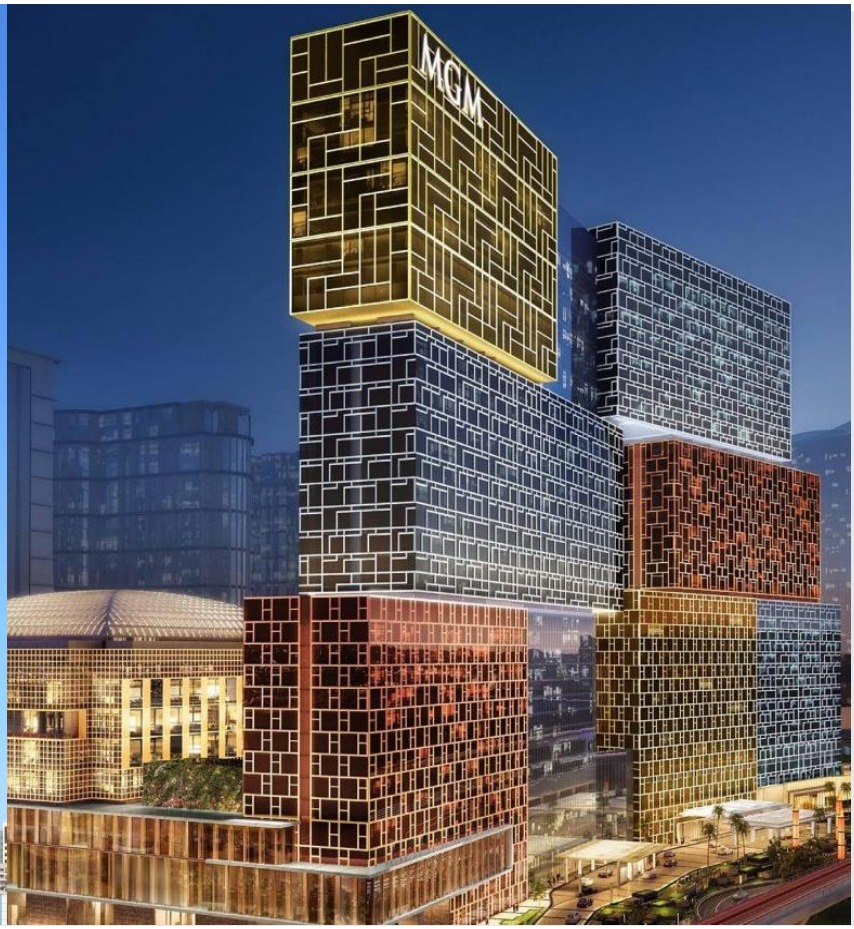
Utilities	2023	
Electricity (kWh)	CDD (Base T=17°C)	
LPG (kg)	Min OAT	
Water (m^3)	CDD (Base T=17°C)	Food Cover

MGM COTAI

Utilities	2023	
Electricity (kWh)	CDD (Base T=17°C)	
Natural Gas (m^3)	Min OAT	
Water (m^3)	CDD (Base T=17°C)	Food Cover

MGM MACAU & COTAI
Resource Efficiency Management Report
860720-0006650.00
860720-0006651.00





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